9 Estimation and Confidence Intervals

© Jack Hollingsworth/Photodisc/Getty Images

THE AMERICAN RESTAURANT ASSOCIATION collected information on the number of meals eaten outside the home per week by young married couples. A survey of 60 couples showed the sample mean number of meals eaten outside the home was 2.76 meals per week, with a standard deviation of 0.75 meal per week. Construct a 99% confidence interval for the population mean. (See Exercise 36 and LO9-2.)

LEARNING OBJECTIVES

When you have completed this chapter, you will be able to:

LO9-1 Compute and interpret a point estimate of a population mean.

LO9-2 Compute and interpret a confidence interval for a population mean.

LO9-3 Compute and interpret a confidence interval for a population proportion.

LO9-4 Calculate the required sample size to estimate a population proportion or population mean.

LO9-5 Adjust a confidence interval for finite populations.

INTRODUCTION

The previous chapter began our discussion of sampling. We introduced both the reasons for, and the methods of, sampling. The reasons for sampling were:

- Contacting the entire population is too time-consuming.
- Studying all the items in the population is often too expensive.
- The sample results are usually adequate.
- Certain tests are destructive.
- Checking all the items is physically impossible.

There are several methods of sampling. Simple random sampling is the most widely used method. With this type of sampling, each member of the population has the same chance of being selected to be a part of the sample. Other methods of sampling include systematic sampling, stratified sampling, and cluster sampling.

Chapter 8 assumes information about the population, such as the mean, the standard deviation, or the shape of the population, is known. In most business situations, such information is not available. In fact, one purpose of sampling is to estimate some of these values. For example, you select a sample from a population and use the mean of the sample to estimate the mean of the population.

This chapter considers several important aspects of sampling. We begin by studying **point estimates.** A point estimate is a single value (point) computed from sample information and used to estimate a population value. For example, we may be interested in the number of hours worked by consultants employed by Boston Consulting Group. Using simple random sampling, we select 50 consultants and ask each of them how many hours they worked last week. The sample's mean is a point estimate of the unknown population mean. A more informative approach is to present a range of values where we expect the population parameter to occur. Such a range of values is called a **confidence interval.**

Frequently in business we need to determine the size of a sample. How many voters should a polling organization contact to forecast the election outcome? How many products do we need to examine to ensure our quality level? This chapter also develops a strategy for determining the appropriate number of observations in the sample.

STATISTICS IN ACTION

On all new cars, a fuel economy estimate is prominently displayed on the window sticker as required by the Environmental Protection Agency (EPA). Often, fuel economy is a factor in a consumer's choice of a new car because of fuel costs or environmental concerns. The fuel estimates for a 2016 BMW 328i Sedan (4-cylinder, automatic) are 23 miles per gallon (mpg) in the city and 35 on the highway. The EPA recognizes that actual fuel economy may differ from the estimates by noting, "No test can simulate all possible combinations of conditions and climate, driver behavior, and car care habits. Actual mileage depends on how, when, and where the vehicle is driven. The EPA has found that the mpg obtained by most drivers will be within a few mpg of the estimates."

POINT ESTIMATE FOR A POPULATION MEAN

LO9-1 Compute and interpret a point estimate of a population mean.

A point estimate is a single statistic used to estimate a population parameter. Suppose Best Buy Inc. wants to estimate the mean age of people who purchase LCD HDTV televisions. They select a random sample of 75 recent purchases, determine the age of each buyer, and compute the mean age of the buyers in the sample. The mean of this sample is a **point estimate** of the population mean.

> **POINT ESTIMATE** The statistic, computed from sample information, that estimates a population parameter.

The following examples illustrate point estimates of population means.

1. Tourism is a major source of income for many Caribbean countries, such as Barbados. Suppose the Bureau of Tourism for Barbados wants an estimate of the mean amount spent by tourists visiting the country. It would not be feasible to contact each tourist. Therefore, 500 tourists are randomly selected as they depart the country and asked in detail about their spending while visiting Barbados. The mean amount spent by the sample of 500 tourists is an estimate of the unknown population parameter. That is, we let the sample mean serve as a point estimate of the population mean.

2. Litchfield Home Builders Inc. builds homes in the southeastern region of the United States. One of the major concerns of new buyers is the date when the home will be completed. Recently, Litchfield has been telling customers, "Your home will be completed 45 working days from the date we begin installing drywall." The customer relations department at Litchfield wishes to compare this pledge with recent experience. A sample of 50 homes completed this year revealed that the point estimate of the population mean is 46.7 working days from the start of drywall to the completion of the home. Is it reasonable to conclude that the population mean is still 45 days and that the difference between the sample mean (46.7 days) and the proposed population mean (45 days) is sampling error? In other words, is the sample mean significantly different from the population mean?
3. Recent medical studies indicate that exercise is an important part of a person's overall health. The director of human resources at OCF, a large glass manufacturer, wants an estimate of the number of hours per week employees spend exercising. A sample of 70 employees reveals the mean number of hours of exercise last week is 3.3. This value is a point estimate of the unknown population mean.


© Andersen Ross/Getty Images RF

The sample mean, $\bar{x}$, is not the only point estimate of a population parameter. For example, p, a sample proportion, is a point estimate of π, the population proportion; and s, the sample standard deviation, is a point estimate of σ, the population standard deviation.

LO9-2
Compute and interpret a confidence interval for a population mean.

CONFIDENCE INTERVALS FOR A POPULATION MEAN

A point estimate, however, tells only part of the story. While we expect the point estimate to be close to the population parameter, we would like to measure how close it really is. A confidence interval serves this purpose. For example, we estimate the mean yearly income for construction workers in the New York–New Jersey area is $85,000. The range of this estimate might be from $81,000 to $89,000. We can describe how confident we are that the population parameter is in the interval. We might say, for instance, that we are 90% confident that the mean yearly income of construction workers in the New York–New Jersey area is between $81,000 and $89,000.

CONFIDENCE INTERVAL A range of values constructed from sample data so that the population parameter is likely to occur within that range at a specified probability. The specified probability is called the *level of confidence*.

To compute a confidence interval for a population mean, we will consider two situations:

- We use sample data to estimate μ with $\bar{x}$ and the population standard deviation (σ) is known.
- We use sample data to estimate μ with $\bar{x}$ and the population standard deviation is unknown. In this case, we substitute the sample standard deviation (s) for the population standard deviation (σ).

There are important distinctions in the assumptions between these two situations. We first consider the case where σ is known.

Population Standard Deviation, Known σ

A confidence interval is computed using two statistics: the sample mean, $\bar{x}$, and the standard deviation. From previous chapters, you know that the standard deviation is an important statistic because it measures the dispersion, or variation, of a population or sampling distribution. In computing a confidence interval, the standard deviation is used to compute the limits of the confidence interval.

To demonstrate the idea of a confidence interval, we start with one simplifying assumption. That assumption is that we know the value of the population standard deviation, σ. Typically, we know the population standard deviation in situations where we have a long history of collected data. Examples are data from monitoring processes that fill soda bottles or cereal boxes, and the results of the SAT Reasoning Test (for college admission). Knowing σ allows us to simplify the development of a confidence interval because we can use the standard normal distribution from Chapter 8.

Recall that the sampling distribution of the sample mean is the distribution of all sample means, $\bar{x}$, of sample size n from a population. The population standard deviation, σ, is known. From this information, and the central limit theorem, we know that the sampling distribution follows the normal probability distribution with a mean of μ and a standard deviation $\sigma/\sqrt{n}$. Also recall that this value is called the standard error.

The results of the central limit theorem allow us to make the following general confidence interval statements using z-statistics:

1. Ninety-five percent of all confidence intervals computed from random samples selected from a population will contain the population mean. These intervals are computed using a z-statistic equal to 1.96.
2. Ninety percent of all confidence intervals computed from random samples selected from a population will contain the population mean. These confidence intervals are computed using a z-statistic equal to 1.65.

These confidence interval statements provide examples of *levels of confidence* and are called a **95% confidence interval** and a **90% confidence interval.** The *95%* and *90%* are the levels of confidence and refer to the percentage of similarly constructed intervals that would include the parameter being estimated—in this case, μ, the population mean.

How are the values of 1.96 and 1.65 obtained? First, let's look for the z value for a 95% confidence interval. The following diagram and Table 9–1 will help explain. Table 9–1 is a reproduction of the standard normal table in Appendix B. However, many rows and columns have been eliminated to allow us to better focus on particular rows and columns.

1. First, we divide the confidence level in half, so .9500/2 = .4750.
2. Next, we find the value .4750 in the body of Table 9–1. Note that .4750 is located in the table at the intersection of a row and a column.
3. Locate the corresponding row value in the left margin, which is 1.9, and the column value in the top margin, which is .06. Adding the row and column values gives us a z value of 1.96.
4. Thus, the probability of finding a z value between 0 and 1.96 is .4750.
5. Likewise, because the normal distribution is symmetric, the probability of finding a z value between −1.96 and 0 is also .4750.
6. When we add these two probabilities, the probability that a z value is between −1.96 and 1.96 is .9500.

For the 90% level of confidence, we follow the same steps. First, one-half of the desired confidence interval is .4500. A search of Table 9–1 does not reveal this exact value. However, it is between two values, .4495 and .4505. As in step three, we locate each value in the table. The first, .4495, corresponds to a z value of 1.64 and the second, .4505, corresponds to a z value of 1.65. To be conservative, we will select the larger of the two z values, 1.65, and the exact level of confidence is 90.1%, or 2(0.4505). Next, the probability of finding a z value between −1.65 and 0 is .4505, and the probability that a z value is between −1.65 and 1.65 is .9010.

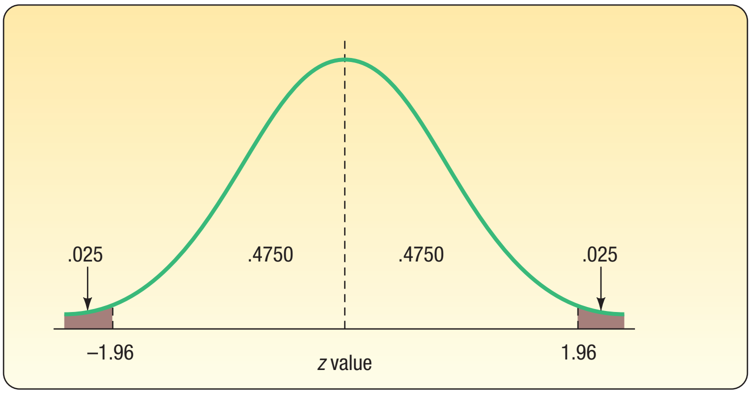


TABLE 9–1 The Standard Normal Table for Selected Values

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884

How do we determine a 95% confidence interval? The width of the interval is determined by two factors: (1) the level of confidence, as described in the previous section, and (2) the size of the standard error of the mean. To find the standard error of the mean, recall from the previous chapter [see formula (8–1) on page 271] that the standard error of the mean reports the variation in the distribution of sample means. It is really the standard deviation of the distribution of sample means. The formula is repeated below:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

where:

- $\sigma_{\bar{x}}$ is the symbol for the standard error of the mean. We use a Greek letter because it is a population value, and the subscript $\bar{x}$ reminds us that it refers to a sampling distribution of the sample means.
- σ is the population standard deviation.
- n is the number of observations in the sample.

The size of the standard error is affected by two values. The first is the standard deviation of the population. The larger the population standard deviation, σ, the larger $\sigma/\sqrt{n}$. If the population is homogeneous, resulting in a small population standard deviation, the standard error will also be small. However, the standard error is also affected by the number of observations in the sample. A large number of observations in the sample will result in a small standard error of estimate, indicating that there is less variability in the sample means.

We can summarize the calculation for a 95% confidence interval using the following formula:

$$\bar{x} \pm 1.96\frac{\sigma}{\sqrt{n}}$$

Similarly, a 90.1% confidence interval is computed as follows:

$$\bar{x} \pm 1.65 \frac{\sigma}{\sqrt{n}}$$

The values 1.96 and 1.65 are z values corresponding to the 95% and the 90.1% confidence intervals, respectively. However, we are not restricted to these values. We can select any confidence level between 0 and 100% and find the corresponding value for z. In general, a confidence interval for the population mean when the population follows the normal distribution and the population standard deviation is known is computed by:

CONFIDENCE INTERVAL FOR A POPULATION MEAN WITH σ KNOWN

$$\bar{x} \pm z \frac{\sigma}{\sqrt{n}} \qquad \textbf{(9–1)}$$

Courtesy Del Monte Corporation

To explain these ideas, consider the following example. Del Monte Foods distributes diced peaches in 4.5-ounce plastic cups. To ensure that each cup contains at least the required amount, Del Monte sets the filling operation to dispense 4.51 ounces of peaches and gel in each cup. Of course, not every cup will contain exactly 4.51 ounces of peaches and gel. Some cups will have more and others less. From historical data, Del Monte knows that 0.04 ounce is the standard deviation of the filling process and that the amount, in ounces, follows the normal probability distribution. The quality control technician selects a sample of 64 cups at the start of each shift, measures the amount in each cup, computes the mean fill amount, and then develops a 95% confidence interval for the population mean. Using the confidence interval, is the process filling the cups to the desired amount? This morning's sample of 64 cups had a sample mean of 4.507 ounces. Based on this information, the 95% confidence interval is:

$$\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}} = 4.507 \pm 1.96 \frac{0.04}{\sqrt{64}} = 4.507 \pm 0.0098$$

The 95% confidence interval estimates that the population mean is between 4.4972 ounces and 4.5168 ounces of peaches and gel. Recall that the process is set to fill each cup with 4.51 ounces. Because the desired fill amount of 4.51 ounces is in this interval, we conclude that the filling process is achieving the desired results. In other words, it is reasonable to conclude that the sample mean of 4.507 could have come from a population distribution with a mean of 4.51 ounces.

In this example, we observe that the population mean of 4.51 ounces is in the confidence interval. But this is not always the case. If we selected 100 samples of 64 cups from the population, calculated the sample mean, and developed a confidence interval based on each sample, we would expect to find the population mean in about 95 of the 100 intervals. Or, in contrast, about five of the intervals would not contain the population mean. From Chapter 8, this is called sampling error. The following example details repeated sampling from a population.

EXAMPLE

The American Management Association (AMA) is studying the income of store managers in the retail industry. A random sample of 49 managers reveals a sample mean of $45,420. The standard deviation of this population is $2,050. The association would like answers to the following questions:

1. What is the population mean?
2. What is a reasonable range of values for the population mean?
3. How do we interpret these results?

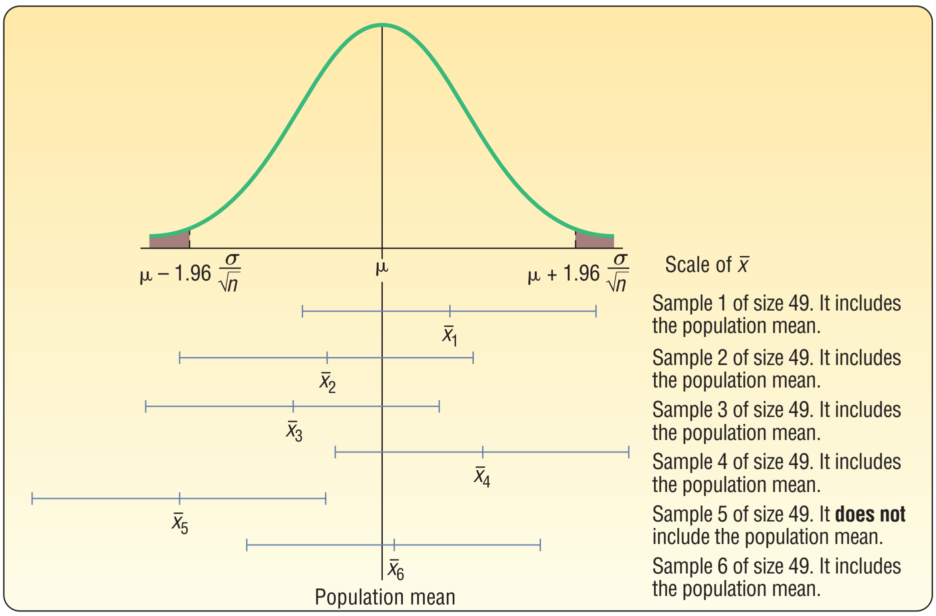
SOLUTION

Generally, distributions of salary and income are positively skewed because a few individuals earn considerably more than others, thus skewing the distribution in the positive direction. Fortunately, the central limit theorem states that the sampling distribution of the mean becomes a normal distribution as sample size increases. In this instance, a sample of 49 store managers is large enough that we can assume that the sampling distribution will follow the normal distribution. Now to answer the questions posed in the example.

1. **What is the population mean?** In this case, we do not know. We do know the sample mean is $45,420. Hence, our best estimate of the unknown population value is the corresponding sample statistic. Thus, the sample mean of $45,420 is a *point estimate* of the unknown population mean.
2. **What is a reasonable range of values for the population mean?** The AMA decides to use the 95% level of confidence. To determine the corresponding confidence interval, we use formula (9–1):

$$\bar{x} \pm z\frac{\sigma}{\sqrt{n}} = \$45{,}420 \pm 1.96\frac{\$2{,}050}{\sqrt{49}} = \$45{,}420 \pm \$574$$

   The confidence interval limits are $44,846 and $45,994 determined by subtracting $574 and adding $574 to the sample mean. The degree or level of confidence is 95% and the confidence interval is from $44,846 to $45,994. The value, $574, is called the margin of error.
3. **How do we interpret these results?** Suppose we select many samples of 49 store managers, perhaps several hundred. For each sample, we compute the mean and then construct a 95% confidence interval, such as we did in the previous section. We could expect about 95% of these confidence intervals to contain the *population* mean. About 5% of the intervals would not contain the population mean annual income, which is μ. However, a particular confidence interval either contains the population parameter or it does not. The following diagram shows the results of selecting samples from the population of store managers in the retail industry, computing the mean of each, and then, using formula (9–1), determining a 95% confidence interval for the population mean. Note that not all intervals include the population mean. Both the endpoints of the fifth sample are less than the population mean. We attribute this to sampling error, and it is the risk we assume when we select the level of confidence.



A Computer Simulation

With statistical software, we can create random samples of a desired sample size, *n*, from a population. For each sample of *n* observations with corresponding numerical values, we can calculate the sample mean. With the sample mean, population standard deviation, and confidence level, we can determine the confidence interval for each sample. Then, using all samples and the confidence intervals, we can find the frequency that the population mean is included in the confidence intervals. The following example does just that.

EXAMPLE

From many years in the automobile leasing business, Town Bank knows that the mean distance driven on an automobile with a four-year lease is 50,000 miles and the standard deviation is 5,000 miles. These are population values. Suppose Town Bank would like to experiment with the idea of sampling to estimate the population mean of 50,000 miles. Town Bank decides to choose a sample size of 30 observations and a 95% confidence interval to estimate the population mean. Based on the experiment, we want to count the number of confidence intervals that include the population mean of 50,000. We expect about 95%, or 57 of the 60 intervals, will include the population mean. To make the calculations easier to understand, we'll conduct the study in thousands of miles, instead of miles.

SOLUTION

Using statistical software, 60 random samples of 30 observations, $n = 30$, are generated and the sample means for each sample computed. Then, using the n of 30 and a standard error of 0.913 ($\sigma/\sqrt{n} = 5/\sqrt{30}$), a 95% confidence interval is computed for each sample. The results of the experiment are shown next.

	Sample Observations													Sample	95% Confidence Limits	
Sample	1	2	3	4	5	–	–	–	26	27	28	29	30	Mean	Lower Limit	Upper Limit
1	56	47	47	48	58	–	–	–	55	62	48	61	57	51.6	49.811	53.389
2	55	51	52	40	53	–	–	–	47	54	55	55	45	50.77	48.981	52.559
3	42	46	48	46	41	–	–	–	50	52	50	47	45	48.63	46.841	50.419
4	52	49	55	47	49	–	–	–	46	56	49	43	50	49.9	48.111	51.689
5	48	50	53	48	45	–	–	–	46	51	61	49	47	49.03	47.241	50.819
6	49	44	47	46	48	–	–	–	51	44	51	52	43	47.73	45.941	49.519
7	50	53	39	50	46	–	–	–	55	47	43	50	57	50.2	48.411	51.989
8	47	51	49	58	44	–	–	–	49	57	54	48	48	51.17	49.381	52.959
9	51	44	47	56	45	–	–	–	45	51	49	49	52	50.33	48.541	52.119
10	45	44	52	52	56	–	–	–	52	51	52	50	48	50	48.211	51.789
11	43	52	54	46	54	–	–	–	43	46	49	52	52	51.2	49.411	52.989
12	57	53	48	42	55	–	–	–	49	44	46	46	48	49.8	48.011	51.589
13	53	39	47	51	53	–	–	–	42	44	44	55	58	49.6	47.811	51.389
14	56	55	45	43	57	–	–	–	48	51	52	55	47	49.03	47.241	50.819
15	49	50	39	45	44	–	–	–	49	43	44	51	51	49.37	47.581	51.159
16	46	44	55	53	55	–	–	–	44	53	53	43	44	50.13	48.341	51.919
17	64	52	55	55	43	–	–	–	58	46	52	58	55	52.47	50.681	54.259
18	57	51	60	40	53	–	–	–	50	51	53	46	52	50.1	48.311	51.889
19	50	49	51	57	45	–	–	–	53	52	40	45	52	49.6	47.811	51.389
20	45	46	53	57	49	–	–	–	49	43	43	53	48	49.47	47.681	51.259
21	52	45	51	52	45	–	–	–	43	49	49	58	53	50.43	48.641	52.219
22	48	48	52	49	40	–	–	–	50	47	54	51	45	47.53	45.741	49.319

(continued)

23	48	50	50	53	44	–	–	–	48	57	52	44	39	49.1	47.311	50.889
24	51	51	40	54	52	–	–	–	54	45	50	57	48	50.13	48.341	51.919
25	48	63	41	52	41	–	–	–	48	50	48	44	53	49.33	47.541	51.119
26	47	45	48	59	49	–	–	–	44	47	49	55	42	49.63	47.841	51.419
27	52	45	60	51	52	–	–	–	52	50	54	46	52	49.4	47.611	51.189
28	46	48	46	57	51	–	–	–	51	50	51	41	52	49.33	47.541	51.119
29	46	48	45	42	48	–	–	–	49	43	59	46	50	48.27	46.481	50.059
30	55	48	47	48	48	–	–	–	47	59	54	51	42	50.53	48.741	52.319
31	58	49	56	46	46	–	–	–	44	51	47	51	46	50.77	48.981	52.559
32	53	54	52	58	55	–	–	–	53	52	45	44	51	50	48.211	51.789
33	50	57	56	51	51	–	–	–	58	47	50	56	46	49.7	47.911	51.489
34	61	48	49	53	54	–	–	–	46	46	56	45	54	50.03	48.241	51.819
35	43	42	43	46	49	–	–	–	49	49	56	51	45	49.43	47.641	51.219
36	39	48	48	51	44	–	–	–	54	52	47	50	52	50.07	48.281	51.859
37	48	43	57	42	54	–	–	–	52	50	59	50	52	50.17	48.381	51.959
38	55	43	49	57	45	–	–	–	41	51	51	52	52	49.5	47.711	51.289
39	47	49	58	54	54	–	–	–	50	56	51	56	58	50.37	48.581	52.159
40	47	56	41	50	54	–	–	–	46	56	61	61	45	51.6	49.811	53.389
41	48	47	42	47	62	–	–	–	44	47	49	55	43	49.43	47.641	51.219
42	46	49	43	36	52	–	–	–	45	51	46	51	43	47.67	45.881	49.459
43	44	48	49	48	51	–	–	–	47	52	51	48	49	49.63	47.841	51.419
44	45	52	54	54	49	–	–	–	49	45	53	50	52	49.07	47.281	50.859
45	54	46	54	45	48	–	–	–	55	38	56	50	62	49.53	47.741	51.319
46	48	50	49	52	51	–	–	–	53	57	58	46	50	49.9	48.111	51.689
47	54	55	46	55	50	–	–	–	56	54	50	55	51	50.5	48.711	52.289
48	45	47	47	63	44	–	–	–	45	53	42	53	50	50.1	48.311	51.889
49	47	47	48	54	56	–	–	–	50	48	54	49	51	49.93	48.141	51.719
50	45	61	51	45	54	–	–	–	55	52	47	45	53	51.03	49.241	52.819
51	49	62	43	49	48	–	–	–	49	58	42	58	52	51.07	49.281	52.859
52	54	52	62	43	54	–	–	–	51	57	49	58	55	50.17	48.381	51.959
53	46	50	59	56	46	–	–	–	50	51	52	54	53	50.47	48.681	52.259
54	52	50	48	48	58	–	–	–	58	52	43	61	54	51.77	49.981	53.559
55	45	44	46	56	46	–	–	–	43	45	63	48	56	49.37	47.581	51.159
56	60	50	56	51	43	–	–	–	45	43	49	59	54	50.37	48.581	52.159
57	59	56	43	47	52	–	–	–	49	54	50	50	57	49.53	47.741	51.319
58	52	55	48	51	40	–	–	–	53	51	51	52	47	49.77	47.981	51.559
59	53	50	44	53	52	–	–	–	47	50	55	46	51	50.07	48.281	51.859
60	55	54	50	52	43	–	–	–	57	50	48	47	53	52.07	50.281	53.859

To explain, in the first row, the statistical software computed 30 random observations from a population distribution with a mean of 50 and a standard deviation of 5. To conserve space, only observations 1 through 5 and 26 through 30 are listed. The first sample's mean is computed and listed as 51.6. In the next columns, the upper and lower limits of the 95% confidence interval for the first sample are shown. The confidence interval calculation for the first sample follows:

$$\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}} = 51.6 \pm 1.96 \frac{5}{\sqrt{30}} = 51.6 \pm 1.789$$

This calculation is repeated for all samples. The results of the experiment show that 93.33%, or 56 of the sixty confidence intervals include the population mean of 50. 93.33% is close to the estimate that 95%, or 57, of the intervals will include the population mean. Using the complement, we expected 5%, or three, of the intervals would not include the population mean. The experiment resulted in 6.67%, or four, of the 60 intervals that did not include the population mean. The particular intervals, 6, 17, 22, and 42, are highlighted in yellow. This is another example of sampling error, or the possibility that a particular random sample may not be a good representation of the population. In each of these four samples, the mean of the sample is either much less or much more than the population mean. Because of random sampling, the mean of the sample is not a good estimate of the population mean, and the confidence interval based on the sample's mean does not include the population mean.

SELF-REVIEW 9–1

The Bun-and-Run is a franchise fast-food restaurant located in the Northeast specializing in half-pound hamburgers, fish sandwiches, and chicken sandwiches. Soft drinks and French fries also are available. The Marketing Department of Bun-and-Run Inc. reports that the distribution of daily sales for their restaurants follows the normal distribution and that the population standard deviation is \$3,000. A sample of 40 franchises showed the mean daily sales to be \$20,000.

(a) What is the population mean of daily sales for Bun-and-Run franchises?
(b) What is the best estimate of the population mean? What is this value called?
(c) Develop a 95% confidence interval for the population mean of daily sales.
(d) Interpret the confidence interval.

EXERCISES

1. A sample of 49 observations is taken from a normal population with a standard deviation of 10. The sample mean is 55. Determine the 99% confidence interval for the population mean.

2. A sample of 81 observations is taken from a normal population with a standard deviation of 5. The sample mean is 40. Determine the 95% confidence interval for the population mean.

3. A sample of 250 observations is selected from a normal population with a population standard deviation of 25. The sample mean is 20.
  a. Determine the standard error of the mean.
  b. Explain why we can use formula (9–1) to determine the 95% confidence interval.
  c. Determine the 95% confidence interval for the population mean.

4. Suppose you know σ and you want an 85% confidence level. What value would you use as z in formula (9–1)?

5. A research firm conducted a survey to determine the mean amount Americans spend on coffee during a week. They found the distribution of weekly spending followed the normal distribution with a population standard deviation of \$5. A sample of 49 Americans revealed that $\bar{x} = \$20$.
  a. What is the point estimate of the population mean? Explain what it indicates.
  b. Using the 95% level of confidence, determine the confidence interval for μ. Explain what it indicates.

6. Refer to the previous exercise. Instead of 49, suppose that 64 Americans were surveyed about their weekly expenditures on coffee. Assume the sample mean remained the same.
  a. What is the 95% confidence interval estimate of μ?
  b. Explain why this confidence interval is narrower than the one determined in the previous exercise.

7. Bob Nale is the owner of Nale's Quick Fill. Bob would like to estimate the mean number of gallons of gasoline sold to his customers. Assume the number of gallons sold follows the normal distribution with a population standard deviation of 2.30 gallons. From his records, he selects a random sample of 60 sales and finds the mean number of gallons sold is 8.60.
  a. What is the point estimate of the population mean?
  b. Develop a 99% confidence interval for the population mean.
  c. Interpret the meaning of part (b).

8. Dr. Patton is a professor of English. Recently she counted the number of misspelled words in a group of student essays. She noted the distribution of misspelled words per essay followed the normal distribution with a population standard deviation of 2.44 words per essay. For her 10 a.m. section of 40 students, the mean number of misspelled words was 6.05. Construct a 95% confidence interval for the mean number of misspelled words in the population of student essays.

Population Standard Deviation, σ Unknown

In the previous section, we assumed the population standard deviation was known. In the case involving Del Monte 4.5-ounce cups of peaches, there would likely be a long history of measurements in the filling process. Therefore, it is reasonable to assume the standard deviation of the population is available. However, in most sampling situations the population standard deviation (σ) is not known. Here are some examples where we wish to estimate the population means and it is unlikely we would know the population standard deviations. Suppose each of these studies involves students at West Virginia University.

- The Dean of the Business College wants to estimate the mean number of hours full-time students work at paying jobs each week. He selects a sample of 30 students, contacts each student, and asks them how many hours they worked last week. From the sample information, he can calculate the sample mean, but it is not likely he would know or be able to find the *population* standard deviation (σ) required in formula (9–1).
- The Dean of Students wants to estimate the distance the typical commuter student travels to class. She selects a sample of 40 commuter students, contacts each, and determines the one-way distance from each student's home to the center of campus. From the sample data, she calculates the mean travel distance, that is, $\bar{x}$. It is unlikely the standard deviation of the population would be known or available, again making formula (9–1) unusable.
- The Director of Student Loans wants to estimate the mean amount owed on student loans at the time of his/her graduation. The director selects a sample of 20 graduating students and contacts each to find the information. From the sample information, the director can estimate the mean amount. However, to develop a confidence interval using formula (9–1), the population standard deviation is necessary. It is not likely this information is available.

STATISTICS IN ACTION

The *t* distribution was created by William Gosset who was born in England in 1876 and died there in 1937. He worked for many years at Arthur Guinness, Sons and Company. In fact, in his later years he was in charge of the Guinness Brewery in London. Guinness preferred its employees to use pen names when publishing papers, so in 1908, when Gosset wrote "The Probable Error of a Mean," he used the name "Student." In this paper, he first described the properties of the *t* distribution and used it to monitor the brewing process so that the beer met Guinness' quality standards.

Fortunately we can use the sample standard deviation to estimate the population standard deviation. That is, we use *s*, the sample standard deviation, to estimate σ, the population standard deviation. But in doing so, we cannot use formula (9–1). Because we do not know σ, we cannot use the *z* distribution. However, there is a remedy. We use the sample standard deviation and replace the *z* distribution with the *t* distribution.

The *t* distribution is a continuous probability distribution, with many similar characteristics to the *z* distribution. William Gosset, an English brewmaster, was the first to study the *t* distribution. He was particularly concerned with the exact behavior of the distribution of the following statistic:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

where *s* is an estimate of σ. He noticed differences between estimating σ based on *s*, especially when *s* was calculated from a very small sample. The *t* distribution and the standard normal distribution are shown graphically in Chart 9–1. Note particularly that the *t* distribution is flatter, more spread out, than the standard normal distribution. This is because the standard deviation of the *t* distribution is larger than that of the standard normal distribution.

The following characteristics of the *t* distribution are based on the assumption that the population of interest is normal, or nearly normal.

- It is, like the z distribution, a continuous distribution.
- It is, like the z distribution, bell-shaped and symmetrical.
- There is not one t distribution, but rather a family of t distributions. All t distributions have a mean of 0, but their standard deviations differ according to the sample size, n. There is a t distribution for a sample size of 20, another for a sample size of 22, and so on. The standard deviation for a t distribution with 5 observations is larger than for a t distribution with 20 observations.
- The t distribution is more spread out and flatter at the center than the standard normal distribution (see Chart 9–1). As the sample size increases, however, the t distribution approaches the standard normal distribution because the errors in using s to estimate σ decrease with larger samples.

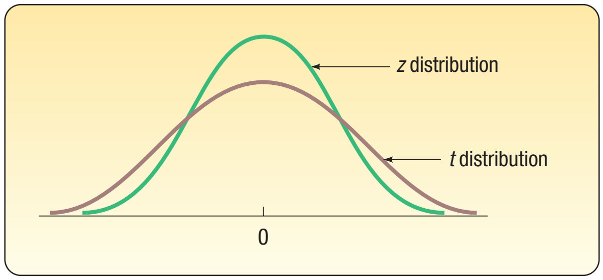


CHART 9–1 The Standard Normal Distribution and Student's t Distribution

Because Student's t distribution has a greater spread than the z distribution, the value of t for a given level of confidence is larger in magnitude than the corresponding z value. Chart 9–2 shows the values of z for a 95% level of confidence and of t for the

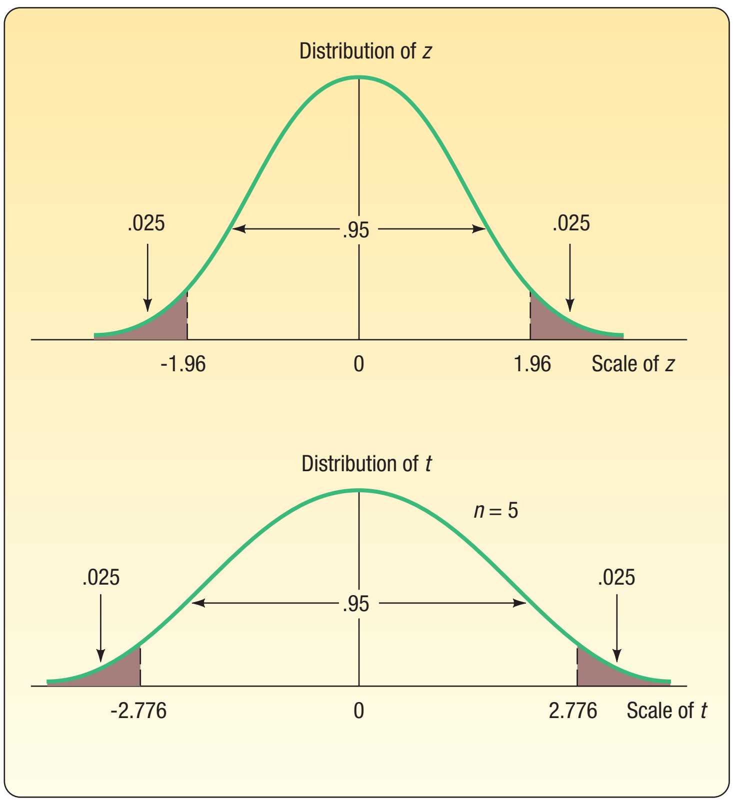


CHART 9–2 Values of z and t for the 95% Level of Confidence

same level of confidence when the sample size is $n = 5$. How we obtained the actual value of t will be explained shortly. For now, observe that for the same level of confidence the t distribution is flatter or more spread out than the standard normal distribution. Note that the 95% confidence interval using a t statistic will be wider compared to an interval using a z statistic.

To develop a confidence interval for the population mean using the t distribution, we adjust formula (9–1) as follows.

CONFIDENCE INTERVAL FOR THE POPULATION MEAN, σ UNKNOWN

$$\bar{x} \pm t\frac{s}{\sqrt{n}} \qquad \textbf{(9–2)}$$

To determine a confidence interval for the population mean with an unknown population standard deviation, we:

1. Assume the sampled population is either normal or approximately normal. This assumption may be questionable for small sample sizes, and becomes more valid with larger sample sizes.
2. Estimate the population standard deviation (σ) with the sample standard deviation (s).
3. Use the t distribution rather than the z distribution.

We should be clear at this point. We base the decision on whether to use the t or the z on whether or not we know σ, the population standard deviation. If we know the population standard deviation, then we use z. If we do not know the population standard deviation, then we must use t. Chart 9–3 summarizes the decision-making process.

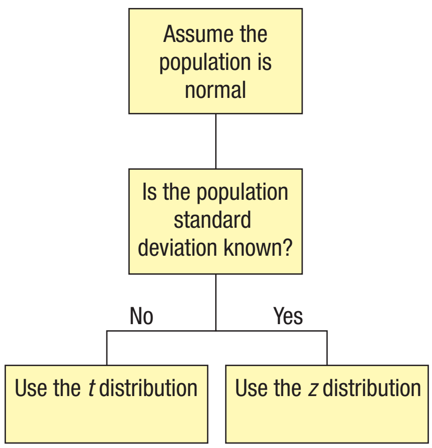


CHART 9–3 Determining When to Use the z Distribution or the t Distribution

The following example will illustrate a confidence interval for a population mean when the population standard deviation is unknown and how to find the appropriate value of t in a table.

EXAMPLE

A tire manufacturer wishes to investigate the tread life of its tires. A sample of 10 tires driven 50,000 miles revealed a sample mean of 0.32 inch of tread remaining with a standard deviation of 0.09 inch. Construct a 95% confidence interval for the population mean. Would it be reasonable for the manufacturer to conclude that after 50,000 miles the population mean amount of tread remaining is 0.30 inch?

SOLUTION

To begin, we assume the population distribution is normal. In this case, we don't have a lot of evidence, but the assumption is probably reasonable. We know the sample standard deviation is .09 inch. We use formula (9–2):

$$\bar{x} \pm t \frac{s}{\sqrt{n}}$$

From the information given, $\bar{x} = 0.32$, $s = 0.09$, and $n = 10$. To find the value of t, we use Appendix B.5, a portion of which is reproduced in Table 9–2. Appendix B.5 is also reproduced on the inside back cover of the text. The first step for locating t is to move across the columns identified for "Confidence Intervals" to the level of confidence requested. In this case, we want the 95% level of confidence, so we move to the column headed "95%." The column on the left margin is identified as "*df.*" This refers to the number of degrees of freedom. The number of degrees of freedom is the number of observations in the sample minus the number of samples, written $n - 1$. In this case, it is $10 - 1 = 9$. Why did we decide there were 9 degrees of freedom? When sample statistics are being used, it is necessary to determine the number of values that are *free to vary.*

TABLE 9–2 A Portion of the *t* Distribution

	Confidence Intervals				
	80%	90%	95%	98%	99%
	Level of Significance for One-Tailed Test				
df	0.10	0.05	0.025	0.010	0.005
	Level of Significance for Two-Tailed Test				
	0.20	0.10	0.05	0.02	0.01
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169

To illustrate the meaning of degrees of freedom: Assume that the mean of four numbers is known to be 5. The four numbers are 7, 4, 1, and 8. The deviations of these numbers from the mean must total 0. The deviations of +2, −1, −4, and +3 do total 0. If the deviations of +2, −1, and −4 are known, then the value of +3 is fixed (restricted) in order to satisfy the condition that the sum of the deviations must equal 0. Thus, 1 degree of freedom is lost in a sampling problem involving the standard deviation of the sample because one number (the arithmetic mean) is known. For a 95% level of confidence and 9 degrees of freedom, we select the row with 9 degrees of freedom. The value of t is 2.262.

To determine the confidence interval, we substitute the values in formula (9–2).

$$\bar{x} \pm t\frac{s}{\sqrt{n}} = 0.32 \pm 2.262\frac{0.09}{\sqrt{10}} = 0.32 \pm 0.64$$

The endpoints of the confidence interval are 0.256 and 0.384. How do we interpret this result? If we repeated this study 200 times, calculating the 95% confidence interval with each sample's mean and the standard deviation, we expect 190 of the intervals would include the population mean. Ten of the intervals would not include the population mean. This is the effect of sampling error. A further interpretation is to conclude that the population mean is in this interval. The manufacturer can be reasonably sure (95% confident) that the mean remaining tread depth is between 0.256 and 0.384 inch. Because the value of 0.30 is in this interval, it is possible that the mean of the population is 0.30.

Here is another example to clarify the use of confidence intervals. Suppose an article in your local newspaper reported that the mean time to sell a residential property in the area is 60 days. You select a random sample of 20 homes sold in the last year and find the mean selling time is 65 days. Based on the sample data, you develop a 95% confidence interval for the population mean. You find that the endpoints of the confidence interval are 62 days and 68 days. How do you interpret this result? You can be reasonably confident the population mean is within this range. The value proposed for the population mean, that is, 60 days, is not included in the interval. It is not likely that the population mean is 60 days. The evidence indicates the statement by the local newspaper may not be correct. To put it another way, it seems unreasonable to obtain the sample you did from a population that had a mean selling time of 60 days.

The following example will show additional details for determining and interpreting a confidence interval. We used Minitab to perform the calculations.

EXAMPLE

The manager of the Inlet Square Mall, near Ft. Myers, Florida, wants to estimate the mean amount spent per shopping visit by customers. A sample of 20 customers reveals the following amounts spent.

$48.16	$42.22	$46.82	$51.45	$23.78	$41.86	$54.86
37.92	52.64	48.59	50.82	46.94	61.83	61.69
49.17	61.46	51.35	52.68	58.84	43.88	

What is the best estimate of the population mean? Determine a 95% confidence interval. Interpret the result. Would it be reasonable to conclude that the population mean is $50? What about $60?

SOLUTION

The mall manager assumes that the population of the amounts spent follows the normal distribution. This is a reasonable assumption in this case. Additionally, the

© McGraw-Hill Education/Andrew Resek, photographer

confidence interval technique is quite powerful and tends to commit any errors on the conservative side if the population is not normal. We should not make the normality assumption when the population is severely skewed or when the distribution has "thick tails." In Chapter 16, we present methods for handling this issue if we cannot make the normality assumption. In this case, the normality assumption is reasonable.

The population standard deviation is not known. Hence, it is appropriate to use the t distribution and formula (9–2) to find the confidence interval. We use the Minitab system to find the mean and standard deviation of this sample. The results are shown below.

Descriptive Statistics: Dollars Spent

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Dollars Spent	20	0	49.348	2.015	9.012	23.780	44.615	49.995	54.315	61.830

1-Sample t: Dollars Spent

Descriptive Statistics

N	Mean	StDev	SE Mean	95% CI for μ
20	49.348	9.012	2.015	(45.130, 53.566)

μ: mean of Dollars Spent

The mall manager does not know the population mean. The sample mean is the best estimate of that value. From the pictured Minitab output, the mean is $49.348, which is the best estimate, the *point estimate,* of the unknown population mean.

We use formula (9–2) to find the confidence interval. The value of t is available from Appendix B.5. There are $n - 1 = 20 - 1 = 19$ degrees of freedom. We move across the row with 19 degrees of freedom to the column for the 95% confidence level. The value at this intersection is 2.093. We substitute these values into formula (9–2) to find the confidence interval.

$$\bar{x} \pm t\frac{s}{\sqrt{n}} = \$49.348 \pm 2.093\frac{\$9.012}{\sqrt{20}} = \$49.348 \pm \$4.218$$

The endpoints of the confidence interval are $45.130 and $53.566. It is reasonable to conclude that the population mean is in that interval.

The manager of Inlet Square wondered whether the population mean could have been $50 or $60. The value of $50 is within the confidence interval. It is reasonable that the population mean could be $50. The value of $60 is not in the confidence interval. Hence, we conclude that the population mean is unlikely to be $60.

The calculations to construct a confidence interval are also available in Excel. The output follows. Note that the sample mean ($49.348) and the sample standard deviation ($9.012) are the same as those in the Minitab calculations. In the Excel output, the last line also includes the margin of error, which is the amount that is added and subtracted from the sample mean to form the endpoints of the confidence interval. This value is found from

$$t\frac{s}{\sqrt{n}} = 2.093\frac{\$9.012}{\sqrt{20}} = \$4.218$$

Amount	
Mean	49.35
Standard Error	2.02
Median	50.00
Mode	#N/A
Standard Deviation	9.01
Sample Variance	81.22
Kurtosis	2.26
Skewness	-1.00
Range	38.05
Minimum	23.78
Maximum	61.83
Sum	986.96
Count	20.00
Confidence Level(95.0%)	4.22

Before doing the confidence interval exercises, we would like to point out a useful characteristic of the t distribution that will allow us to use the t table to quickly find both z and t values. Earlier in this section on page 293, we detailed the characteristics of the t distribution. The last point indicated that as we increase the sample size the t distribution approaches the z distribution. In fact, when we reach an infinitely large sample, the t distribution is exactly equal to the z distribution.

To explain, Table 9–3 is a portion of Appendix B.5, with the degrees of freedom between 4 and 99 omitted. To find the appropriate z value for a 95% confidence interval,

TABLE 9–3 Student's *t* Distribution

df (degrees of freedom)	Confidence Interval					
	80%	**90%**	**95%**	**98%**	**99%**	**99.9%**
	Level of Significance for One-Tailed Test, α					
	0.1	**0.05**	**0.025**	**0.01**	**0.005**	**0.0005**
	Level of Significance for Two-Tailed Test, α					
	0.2	**0.1**	**0.05**	**0.02**	**0.01**	**0.001**
1	3.078	6.314	12.706	31.821	63.657	636.619
2	1.886	2.920	4.303	6.965	9.925	31.599
3	1.638	2.353	3.182	4.541	5.841	12.924
⋮	⋮	⋮	⋮	⋮	⋮	⋮
100	1.290	1.660	1.984	2.364	2.626	3.390
120	1.289	1.658	1.980	2.358	2.617	3.373
140	1.288	1.656	1.977	2.353	2.611	3.361
160	1.287	1.654	1.975	2.350	2.607	3.352
180	1.286	1.653	1.973	2.347	2.603	3.345
200	1.286	1.653	1.972	2.345	2.601	3.340
∞	1.282	1.645	1.960	2.326	2.576	3.291

we begin by going to the confidence interval section and selecting the column headed "95%." Move down that column to the last row, which is labeled "∞," or infinite degrees of freedom. The value reported is 1.960, the same value that we found using the standard normal distribution in Appendix B.3. This confirms the convergence of the t distribution to the z distribution.

What does this mean for us? Instead of searching in the body of the z table, we can go to the last row of the t table and find the appropriate value to build a confidence interval. An additional benefit is that the values have three decimal places. So, using this table for a 90% confidence interval, go down the column headed "90%" and see the value 1.645, which is a more precise z value that can be used for the 90% confidence level. Other z values for 98% and 99% confidence intervals are also available with three decimals. Note that we will use the t table, which is summarized in Table 9–3, to find the z values with three decimals for all following exercises and problems.

SELF-REVIEW 9–2

Dottie Kleman is the "Cookie Lady." She bakes and sells cookies at locations in the Philadelphia area. Ms. Kleman is concerned about absenteeism among her workers. The information below reports the number of days absent for a sample of 10 workers during the last two-week pay period.

4	1	2	2	1	2	2	1	0	3

(a) Determine the mean and the standard deviation of the sample.
(b) What is the population mean? What is the best estimate of that value?
(c) Develop a 95% confidence interval for the population mean. Assume that the population distribution is normal.
(d) Explain why the t distribution is used as a part of the confidence interval.
(e) Is it reasonable to conclude that the typical worker does not miss any days during a pay period?

EXERCISES

9. Use Appendix B.5 to locate the value of t under the following conditions.
 a. The sample size is 12 and the level of confidence is 95%.
 b. The sample size is 20 and the level of confidence is 90%.
 c. The sample size is 8 and the level of confidence is 99%.

10. Use Appendix B.5 to locate the value of t under the following conditions.
 a. The sample size is 15 and the level of confidence is 95%.
 b. The sample size is 24 and the level of confidence is 98%.
 c. The sample size is 12 and the level of confidence is 90%.

11. The owner of Britten's Egg Farm wants to estimate the mean number of eggs produced per chicken. A sample of 20 chickens shows they produced an average of 20 eggs per month with a standard deviation of 2 eggs per month.
 a. What is the value of the population mean? What is the best estimate of this value?
 b. Explain why we need to use the t distribution. What assumption do you need to make?
 c. For a 95% confidence interval, what is the value of t?
 d. Develop the 95% confidence interval for the population mean.
 e. Would it be reasonable to conclude that the population mean is 21 eggs? What about 25 eggs?

12. The U.S. Dairy Industry wants to estimate the mean yearly milk consumption. A sample of 16 people reveals the mean yearly consumption to be 45 gallons with a standard deviation of 20 gallons. Assume the population distribution is normal.

a. What is the value of the population mean? What is the best estimate of this value?
b. Explain why we need to use the *t* distribution. What assumption do you need to make?
c. For a 90% confidence interval, what is the value of *t*?
d. Develop the 90% confidence interval for the population mean.
e. Would it be reasonable to conclude that the population mean is 48 gallons?

13. **FILE** Merrill Lynch Securities and Health Care Retirement Inc. are two large employers in downtown Toledo, Ohio. They are considering jointly offering child care for their employees. As a part of the feasibility study, they wish to estimate the mean weekly child-care cost of their employees. A sample of 10 employees who use child care reveals the following amounts spent last week.

\$107	\$92	\$97	\$95	\$105	\$101	\$91	\$99	\$95	\$104

Develop a 90% confidence interval for the population mean. Interpret the result.

14. **FILE** The Columbus, Ohio Area Chamber of Commerce wants to estimate the mean time workers who are employed in the downtown area spend getting to work. A sample of 15 workers reveals the following number of minutes spent traveling.

14	24	24	19	24	7	31	20
26	23	23	28	16	15	21	

Develop a 98% confidence interval for the population mean. Interpret the result.

LO9-3
Compute and interpret a confidence interval for a population proportion.

A CONFIDENCE INTERVAL FOR A POPULATION PROPORTION

The material presented so far in this chapter uses the ratio scale of measurement. That is, we use such variables as incomes, weights, distances, and ages. We now want to consider situations such as the following:

© Vytautas Kielaitis/Shutterstock.com

- The career services director at Southern Technical Institute reports that 80% of its graduates enter the job market in a position related to their field of study.
- A company representative claims that 45% of Burger King sales are made at the drive-through window.
- A survey of homes in the Chicago area indicated that 85% of the new construction had central air conditioning.
- A recent survey of married men between the ages of 35 and 50 found that 63% felt that both partners should earn a living.

These examples illustrate the nominal scale of measurement when the outcome is limited to two values. In these cases, an observation is classified into one of two mutually exclusive groups. For example, a graduate of Southern Tech either entered the job market in a position related to his or her field of study or not. A particular Burger King customer either made a purchase at the drive-through window or did not make a purchase at the drive-through window. We can talk about the groups in terms of proportions.

PROPORTION The fraction, ratio, or percent indicating the part of the sample or the population having a particular trait of interest.

As an example of a proportion, a recent survey indicated that 92 out of 100 people surveyed favored the continued use of daylight savings time in the summer. The sample proportion is 92/100, or .92, or 92%. If we let p represent the sample proportion, x the number of "successes," and n the number of items sampled, we can determine a sample proportion as follows.

SAMPLE PROPORTION

$$p = \frac{x}{n} \tag{9–3}$$

The population proportion is identified by π. Therefore, π refers to the percent of successes in the population. Recall from Chapter 6 that π is the proportion of "successes" in a binomial distribution. This continues our practice of using Greek letters to identify population parameters and Roman letters to identify sample statistics.

To develop a confidence interval for a proportion, we need to meet two requirements:

1. The binomial conditions, discussed in Chapter 6, have been met. These conditions are:
    a. The sample data are the number of successes in n trials.
    b. There are only two possible outcomes. (We usually label one of the outcomes a "success" and the other a "failure.")
    c. The probability of a success remains the same from one trial to the next.
    d. The trials are independent. This means the outcome on one trial does not affect the outcome on another.
2. The values $n\pi$ and $n(1 - \pi)$ should both be greater than or equal to 5. This allows us to invoke the central limit theorem and employ the standard normal distribution, that is, z, to complete a confidence interval.

Developing a point estimate for a population proportion and a confidence interval for a population proportion is similar to doing so for a mean. To illustrate, John Gail is running for Congress from the third district of Nebraska. From a random sample of 100 voters in the district, 60 indicate they plan to vote for him in the upcoming election. The sample proportion is .60, but the population proportion is unknown. That is, we do not know what proportion of voters in the *population* will vote for Mr. Gail. The sample value, .60, is the best estimate we have of the unknown population parameter. So we let p, which is .60, be an estimate of π, which is not known.

To develop a confidence interval for a population proportion, we use:

CONFIDENCE INTERVAL FOR A POPULATION PROPORTION

$$p \pm z\sqrt{\frac{p(1 - p)}{n}} \tag{9–4}$$

An example will help to explain the details of determining a confidence interval and interpreting the result.

EXAMPLE

The union representing the Bottle Blowers of America (BBA) is considering a proposal to merge with the Teamsters Union. According to BBA union bylaws, at least three-fourths of the union membership must approve any merger. A random sample of 2,000 current BBA members reveals 1,600 plan to vote for the merger proposal. What is the estimate of the population proportion? Develop a 95% confidence interval for the population proportion. Basing your decision on this sample information, can you conclude that the necessary proportion of BBA members favor the merger? Why?

SOLUTION

First, calculate the sample proportion from formula (9–3). It is .80, found by

$$p = \frac{x}{n} = \frac{1{,}600}{2{,}000} = .80$$

Thus, we estimate that 80% of the population favor the merger proposal. We determine the 95% confidence interval using formula (9–4). The z value corresponding to the 95% level of confidence is 1.96.

$$p \pm z\sqrt{\frac{p(1-p)}{n}} = .80 \pm 1.96\sqrt{\frac{.80(1-.80)}{2{,}000}} = .80 \pm .018$$

The endpoints of the confidence interval are .782 and .818. The lower endpoint is greater than .75. Hence, we conclude that the merger proposal will likely pass because the interval estimate includes only values greater than 75% of the union membership.

To review the interpretation of the confidence interval: If the poll was conducted 100 times with 100 different samples, we expect the confidence intervals constructed from 95 of the samples would contain the true population proportion. In addition, the interpretation of a confidence interval can be very useful in decision making and play a very important role especially on election night. For example, Cliff Obermeyer is running for Congress from the 6th District of New Jersey. Suppose 500 voters are contacted upon leaving the polls and 275 indicate they voted for Mr. Obermeyer. We will assume that the exit poll of 500 voters is a random sample of those voting in the 6th District. That means that 55% of those in the sample voted for Mr. Obermeyer. Based on formula (9–3):

$$p = \frac{x}{n} = \frac{275}{500} = .55$$

Now, to be assured of election, he must earn *more than* 50% of the votes in the population of those voting. At this point, we know a point estimate, which is .55, of the population of voters that will vote for him. But we do not know the percent in the population that will ultimately vote for the candidate. So the question is: Could we take a sample of 500 voters from a population where 50% or less of the voters support Mr. Obermeyer and find that 55% of the sample support him? To put it another way, could the sampling error, which is $p - \pi = .55 - .50 = .05$ be due to chance, or is the population of voters who support Mr. Obermeyer greater than .50? If we develop a confidence interval for the sample proportion and find that the lower endpoint is greater than .50, then we conclude that the proportion of voters supporting Mr. Obermeyer is greater than .50. What does that mean? Well, it means he should be elected! What if .50 is in the interval? Then we conclude that he is not assured of a majority and we cannot conclude he will be elected. In this case, using the 95% significance level and formula (9–4):

$$p \pm z\sqrt{\frac{p(1-p)}{n}} = .55 \pm 1.96\sqrt{\frac{.55(1-.55)}{500}} = .55 \pm .044$$

So the endpoints of the confidence interval are .55 – .044 = .506 and .55 + .044 = .594. The value of .50 is not in this interval. So we conclude that probably *more than 50%* of the voters support Mr. Obermeyer and that is enough to get him elected.

Is this procedure ever used? Yes! It is exactly the procedure used by polling organizations, television networks, and surveys of public opinion on election night.

STATISTICS IN ACTION

The results of many surveys include confidence intervals. For example, a recent survey of 800 TV viewers in Toledo, Ohio, found 44% watched the evening news on the local CBS affiliate. The article also reported a margin of error of 3.4%. The margin of error is actually the amount that is added and subtracted from the point estimate to find the endpoints of a confidence interval. For a 95% level of confidence, the margin of error is:

$$z\sqrt{\frac{p(1-p)}{n}} = 1.96\sqrt{\frac{.44(1-.44)}{800}} = 0.034$$

The estimate of the proportion of all TV viewers in Toledo, Ohio who watch the local news on CBS is between (.44 – .034) and (.44 + .034) or 40.6% and 47.4%.

SELF-REVIEW 9–3

A market research consultant was hired to estimate the proportion of homemakers who associate the brand name of a laundry detergent with the container's shape and color. The consultant randomly selected 1,400 homemakers. From the sample, 420 were able to identify the brand by name based only on the shape and color of the container.

(a) Estimate the value of the population proportion.
(b) Develop a 99% confidence interval for the population proportion.
(c) Interpret your findings.

EXERCISES

15. The owner of the West End Kwick Fill Gas Station wishes to determine the proportion of customers who pay at the pump using a credit card or debit card. He surveys 100 customers and finds that 80 paid at the pump.
- **a.** Estimate the value of the population proportion.
- **b.** Develop a 95% confidence interval for the population proportion.
- **c.** Interpret your findings.

16. Ms. Maria Wilson is considering running for mayor of Bear Gulch, Montana. Before completing the petitions, she decides to conduct a survey of voters in Bear Gulch. A sample of 400 voters reveals that 300 would support her in the November election.
- **1.** Estimate the value of the population proportion.
- **2.** Develop a 99% confidence interval for the population proportion.
- **3.** Interpret your findings.

17. The Fox TV network is considering replacing one of its prime-time crime investigation shows with a new family-oriented comedy show. Before a final decision is made, network executives designed an experiment to estimate the proportion of their viewers who would prefer the comedy show over the crime investigation show. A random sample of 400 viewers was selected and asked to watch the new comedy show and the crime investigation show. After viewing the shows, 250 indicated they would watch the new comedy show and suggested it replace the crime investigation show.
- **a.** Estimate the value of the population proportion of people who would prefer the comedy show.
- **b.** Develop a 99% confidence interval for the population proportion of people who would prefer the comedy show.
- **c.** Interpret your findings.

18. Schadek Silkscreen Printing Inc. purchases plastic cups and imprints them with logos for sporting events, proms, birthdays, and other special occasions. Zack Schadek, the owner, received a large shipment this morning. To ensure the quality of the shipment, he selected a random sample of 300 cups and inspected them for defects. He found 15 to be defective.
- **a.** What is the estimated proportion defective in the population?
- **b.** Develop a 95% confidence interval for the proportion defective.
- **c.** Zack has an agreement with his supplier that if 10% or more of the cups are defective, he can return the order. Should he return this lot? Explain your decision.

LO9-4
Calculate the required sample size to estimate a population proportion or population mean.

CHOOSING AN APPROPRIATE SAMPLE SIZE

When working with confidence intervals, one important variable is sample size. However, in practice, sample size is not a variable. It is a decision we make so that our estimate of a population parameter is a good one. Our decision is based on three variables:

1. The margin of error the researcher will tolerate.
2. The level of confidence desired, for example, 95%.
3. The variation or dispersion of the population being studied.

The first variable is the *margin of error.* It is designated as E and is the amount that is added and subtracted to the sample mean (or sample proportion) to determine the endpoints of the confidence interval. For example, in a study of wages, we may decide that we want to estimate the mean wage of the population with a margin of error of plus or minus \$1,000. Or, in an opinion poll, we may decide that we want to estimate the population proportion with a margin of error of plus or minus 3.5%. The margin of error is the amount of error we are willing to tolerate in estimating a population parameter. You may wonder why we do not choose small margins of error. There is a trade-off between the margin of error and sample size. A small margin of error will require a larger sample and more money and time to collect the sample. A larger margin of error will permit a smaller sample and result in a wider confidence interval.

The second choice is the *level of confidence.* In working with confidence intervals, we logically choose relatively high levels of confidence such as 95% and 99%. To compute the sample size, we need the z-statistic that corresponds to the chosen level of confidence. The 95% level of confidence corresponds to a z value of 1.96, and a 90% level of confidence corresponds to a z value of 1.645 (using the t table). Notice that larger sample sizes (and more time and money to collect the sample) correspond with higher levels of confidence. Also, notice that we use a z-statistic.

The third choice to determine the sample size is the *population standard deviation.* If the population is widely dispersed, a large sample is required to get a good estimate. On the other hand, if the population is concentrated (homogeneous), the required sample size to get a good estimate will be smaller. Often, we do not know the population standard deviation. Here are three suggestions to estimate the population standard deviation.

1. **Conduct a pilot study.** This is the most common method. Suppose we want an estimate of the number of hours per week worked by students enrolled in the College of Business at the University of Texas. To test the validity of our questionnaire, we use it on a small sample of students. From this small sample, we compute the standard deviation of the number of hours worked and use this value as the population standard deviation.
2. **Use a comparable study.** Use this approach when there is an estimate of the standard deviation from another study. Suppose we want to estimate the number of hours worked per week by refuse workers. Information from certain state or federal agencies that regularly study the workforce may provide a reliable value to use for the population standard deviation.
3. **Use a range-based approach.** To use this approach, we need to know or have an estimate of the largest and smallest values in the population. Recall from Chapter 3, the Empirical Rule states that virtually all the observations could be expected to be within plus or minus 3 standard deviations of the mean, assuming that the distribution follows the normal distribution. Thus, the distance between the largest and the smallest values is 6 standard deviations. We can estimate the standard deviation as one-sixth of the range. For example, the director of operations at University Bank wants to estimate the number of ATM transactions per month made by college students. She believes that the distribution of ATM transactions follows the normal distribution. The minimum and maximum of ATM transactions per month are 2 and 50, so the range is 48, found by (50 − 2). Then the estimated value of the population standard deviation would be eight ATM transactions per month, 48/6.

Sample Size to Estimate a Population Mean

To estimate a population mean, we can express the interaction among these three factors and the sample size in the following formula. Notice that this formula is the margin of error used to calculate the endpoints of confidence intervals to estimate a population mean! See formula 9–1.

$$E = z\frac{\sigma}{\sqrt{n}}$$

Solving this equation for n yields the following result.

SAMPLE SIZE FOR ESTIMATING THE POPULATION MEAN

$$n = \left(\frac{z\sigma}{E}\right)^2 \qquad \textbf{(9–5)}$$

where:

n is the size of the sample.
z is the standard normal z-value corresponding to the desired level of confidence.
σ is the population standard deviation.
E is the maximum allowable error.

The result of this calculation is not always a whole number. When the outcome is not a whole number, the usual practice is to round up *any* fractional result to the next whole number. For example, 201.21 would be rounded up to 202.

EXAMPLE

A student in public administration wants to estimate the mean monthly earnings of city council members in large cities. She can tolerate a margin of error of \$100 in estimating the mean. She would also prefer to report the interval estimate with a 95% level of confidence. The student found a report by the Department of Labor that reported a standard deviation of \$1,000. What is the required sample size?

SOLUTION

The maximum allowable error, E, is \$100. The value of z for a 95% level of confidence is 1.96, and the value of the standard deviation is \$1,000. Substituting these values into formula (9–5) gives the required sample size as:

$$n = \left(\frac{z\sigma}{E}\right)^2 = \left(\frac{(1.96)(\$1{,}000)}{\$100}\right)^2 = (19.6)^2 = 384.16$$

The computed value of 384.16 is rounded up to 385. A sample of 385 is required to meet the specifications. If the student wants to increase the level of confidence, for example to 99%, this will require a larger sample. Using the t table with infinite degrees of freedom, the z value for a 99% level of confidence is 2.576.

$$n = \left(\frac{z\sigma}{E}\right)^2 = \left(\frac{(2.576)(\$1{,}000)}{\$100}\right)^2 = (25.76)^2 = 663.58$$

We recommend a sample of 664. Observe how much the change in the confidence level changed the size of the sample. An increase from the 95% to the 99% level of confidence resulted in an increase of 279 observations, or 72% [(664/385) × 100]. This would greatly increase the cost of the study, in terms of both time and money. Hence, the level of confidence should be considered carefully.

Sample Size to Estimate a Population Proportion

To determine the sample size to estimate a population proportion, the same three variables need to be specified:

1. The margin of error.
2. The desired level of confidence.
3. The variation or dispersion of the population being studied.

For the binomial distribution, the margin of error is:

$$E = z\sqrt{\frac{\pi(1-\pi)}{n}}$$

Solving this equation for n yields the following equation

SAMPLE SIZE FOR THE POPULATION PROPORTION

$$n = \pi(1-\pi)\left(\frac{z}{E}\right)^2 \qquad \textbf{(9–6)}$$

where:

n is the size of the sample.
z is the standard normal z-value corresponding to the desired level of confidence.
π is the population proportion.
E is the maximum allowable error.

As before, the z-value is associated with our choice of confidence level. We also decide the margin of error, E. However, the population variance of the binomial distribution is represented by $\pi(1 - \pi)$. To estimate the population variance, we need a value of the population proportion. If a reliable value cannot be determined with a pilot study or found in a comparable study, then a value of .50 can be used for π. Note that $\pi(1 - \pi)$ has the largest value using 0.50 and, therefore, without a good estimate of the population proportion, using 0.50 as an estimate of π overstates the sample size. Using a larger sample size will not hurt the estimate of the population proportion.

EXAMPLE

The student in the previous example also wants to estimate the proportion of cities that have private refuse collectors. The student wants to estimate the population proportion with a margin of error of .10, prefers a level of confidence of 90%, and has no estimate for the population proportion. What is the required sample size?

SOLUTION

The estimate of the population proportion is to be within .10, so E = .10. The desired level of confidence is .90, which corresponds to a z value of 1.645, using the t table with infinite degrees of freedom. Because no estimate of the population proportion is available, we use .50. The suggested number of observations is

$$n = (.5)(1 - .5)\left(\frac{1.645}{.10}\right)^2 = 67.65$$

The student needs a random sample of 68 cities.

SELF-REVIEW 9–4

A university's office of research wants to estimate the arithmetic mean grade point average (GPA) of all graduating seniors during the past 10 years. GPAs range between 2.0 and 4.0. The estimate of the population mean GPA should be within plus or minus .05 of the population mean. Based on prior experience, the population standard deviation is 0.279. Using a 99% level of confidence, how many student records need to be selected?

EXERCISES

19. A population's standard deviation is 10. We want to estimate the population mean within 2, with a 95% level of confidence. How large a sample is required?

20. We want to estimate the population mean within 5, with a 99% level of confidence. The population standard deviation is estimated to be 15. How large a sample is required?

21. The estimate of the population proportion should be within plus or minus .05, with a 95% level of confidence. The best estimate of the population proportion is .15. How large a sample is required?

22. The estimate of the population proportion should be within plus or minus .10, with a 99% level of confidence. The best estimate of the population proportion is .45. How large a sample is required?

23. A large on-demand, video streaming company is designing a large-scale survey to determine the mean amount of time corporate executives watch on-demand television. A small pilot survey of 10 executives indicated that the mean time per week is 12 hours, with a standard deviation of 3 hours. The estimate of the mean viewing time should be within one-quarter hour. The 95% level of confidence is to be used. How many executives should be surveyed?

24. A processor of carrots cuts the green top off each carrot, washes the carrots, and inserts six to a package. Twenty packages are inserted in a box for shipment. Each box of carrots should weigh 20.4 pounds. The processor knows that the standard deviation of box weight is 0.5 pound. The processor wants to know if the current packing process meets the 20.4 weight standard. How many boxes must the processor sample to be 95% confident that the estimate of the population mean is within 0.2 pound?

25. Suppose the U.S. president wants to estimate the proportion of the population that supports his current policy toward revisions in the health care system. The president wants the estimate to be within .04 of the true proportion. Assume a 95% level of confidence. The president's political advisors found a similar survey from two years ago that reported that 60% of people supported health care revisions.

- **a.** How large of a sample is required?
- **b.** How large of a sample would be necessary if no estimate were available for the proportion supporting current policy?

26. Past surveys reveal that 30% of tourists going to Las Vegas to gamble spend more than $1,000. The Visitor's Bureau of Las Vegas wants to update this percentage.

- **a.** How many tourists should be randomly selected to estimate the population proportion with a 90% confidence level and a 1% margin of error.
- **b.** The Bureau feels the sample size determined above is too large. What can be done to reduce the sample? Based on your suggestion, recalculate the sample size.

LO9-5
Adjust a confidence interval for finite populations.

FINITE-POPULATION CORRECTION FACTOR

The populations we have sampled so far have been very large or infinite. What if the sampled population is not very large? We need to make some adjustments in the way we compute the standard error of the sample means and the standard error of the sample proportions.

A population that has a fixed upper bound is *finite*. For example, there are 7,640 students enrolled at Eastern Illinois University, there are 40 employees at Spence Sprockets, there were 917 Jeep Wranglers assembled at the Alexis Avenue plant yesterday, or there were 65 surgical patients at St. Rose Memorial Hospital in Sarasota yesterday. A finite population can be rather small; it could be all the students registered for your statistics class. It can also be very large, such as all senior citizens living in Florida.

For a finite population, where the total number of objects or individuals is N and the number of objects or individuals in the sample is n, we need to adjust the standard errors in the confidence interval formulas. To put it another way, to find the confidence interval for the mean, we adjust the standard error of the mean in formulas (9–1) and (9–2). If we are determining the confidence interval for a proportion, then we need to adjust the standard error of the proportion in formula (9–4).

This adjustment is called the **finite-population correction factor.** It is often shortened to *FPC* and is:

$$FPC = \sqrt{\frac{N-n}{N-1}}$$

Why is it necessary to apply a factor, and what is its effect? Logically, if the sample is a substantial percentage of the population, the estimate of the population parameter is more precise. Note the effect of the term $(N - n)/(N - 1)$. Suppose the population is 1,000 and the sample is 100. Then this ratio is (1,000 − 100)/(1,000 − 1), or 900/999. Taking the square root gives the correction factor, .9492. Multiplying this correction factor by the standard error *reduces* the standard error by about 5% (1 − .9492 = .0508). This reduction in the size of the standard error yields a smaller range of values in estimating the population mean or the population proportion. If the sample is 200, the correction factor is .8949, meaning that the standard error has been reduced by more than 10%. Table 9–4 shows the effects of various sample sizes.

TABLE 9–4 Finite-Population Correction Factor for Selected Samples When the Population Is 1,000

Sample Size	Fraction of Population	Correction Factor
10	.010	.9955
25	.025	.9879
50	.050	.9752
100	.100	.9492
200	.200	.8949
500	.500	.7075

So if we wished to develop a confidence interval for the mean from a finite population and the population standard deviation was unknown, we would adjust formula (9–2) as follows:

$$\bar{x} \pm t\frac{s}{\sqrt{n}}\left(\sqrt{\frac{N-n}{N-1}}\right)$$

We would make a similar adjustment to formula (9–4) in the case of a proportion.

The following example summarizes the steps to find a confidence interval for the mean.

EXAMPLE

There are 250 families residing in Scandia, Pennsylvania. A random sample of 40 of these families revealed the mean annual church contribution was \$450 and the standard deviation of this was \$75.

1. What is the population mean? What is the best estimate of the population mean?
2. Develop a 90% confidence interval for the population mean. What are the endpoints of the confidence interval?
3. Using the confidence interval, explain why the population mean could be \$445. Could the population mean be \$425? Why?

SOLUTION

First, note the population is finite. That is, there is a limit to the number of people residing in Scandia, in this case 250.

1. We do not know the population mean. This is the value we wish to estimate. The best estimate we have of the population mean is the sample mean, which is \$450.
2. The formula to find the confidence interval for a population mean follows.

$$\bar{x} \pm t\frac{s}{\sqrt{n}}\left(\sqrt{\frac{N-n}{N-1}}\right)$$

   In this case, we know $\bar{x} = 450$, $s = 75$, $N = 250$, and $n = 40$. We do not know the population standard deviation, so we use the t distribution. To find the appropriate value of t, we use Appendix B.5, and move across the top row to the column headed 90%. The degrees of freedom are $df = n - 1 = 40 - 1 = 39$, so we move to the cell where the df row of 39 intersects with the column headed 90%. The value is 1.685. Inserting these values in the formula:

$$\bar{x} \pm t\frac{s}{\sqrt{n}}\left(\sqrt{\frac{N-n}{N-1}}\right)$$

$$= \$450 \pm 1.685\frac{\$75}{\sqrt{40}}\left(\sqrt{\frac{250-40}{250-1}}\right) = \$450 \pm \$19.98\sqrt{.8434} = \$450 \pm \$18.35$$

   The endpoints of the confidence interval are \$431.65 and \$468.35.
3. It is likely that the population mean is more than \$431.65 but less than \$468.35. To put it another way, could the population mean be \$445? Yes, but it is not likely that it is \$425. Why is this so? Because the value \$445 is within the confidence interval and \$425 is not within the confidence interval.

SELF-REVIEW 9–5

The same study of church contributions in Scandia revealed that 15 of the 40 families sampled attend church regularly. Construct the 95% confidence interval for the proportion of families attending church regularly.

EXERCISES

27. Thirty-six items are randomly selected from a population of 300 items. The sample mean is 35 and the sample standard deviation 5. Develop a 95% confidence interval for the population mean.

28. Forty-nine items are randomly selected from a population of 500 items. The sample mean is 40 and the sample standard deviation 9. Develop a 99% confidence interval for the population mean.

29. The attendance at the Savannah Colts minor league baseball game last night was 400. A random sample of 50 of those in attendance revealed that the mean number of soft drinks consumed per person was 1.86, with a standard deviation of 0.50. Develop a 99% confidence interval for the mean number of soft drinks consumed per person.

30. There are 300 welders employed at Maine Shipyards Corporation. A sample of 30 welders revealed that 18 graduated from a registered welding course. Construct the 95% confidence interval for the proportion of all welders who graduated from a registered welding course.

CHAPTER SUMMARY

I. A point estimate is a single value (statistic) used to estimate a population value (parameter).

II. A confidence interval is a range of values within which the population parameter is expected to occur.

A. The factors that determine the width of a confidence interval for a mean are:

1. The number of observations in the sample, *n*.

2. The variability in the population, usually estimated by the sample standard deviation, *s*.

3. The level of confidence.

a. To determine the confidence limits when the population standard deviation is known, we use the *z* distribution. The formula is

$$\bar{x} \pm z \frac{\sigma}{\sqrt{n}} \qquad \textbf{(9–1)}$$

b. To determine the confidence limits when the population standard deviation is unknown, we use the *t* distribution. The formula is

$$\bar{x} \pm t \frac{s}{\sqrt{n}} \qquad \textbf{(9–2)}$$

III. The major characteristics of the *t* distribution are:

A. It is a continuous distribution.

B. It is mound-shaped and symmetrical.

C. It is flatter, or more spread out, than the standard normal distribution.

D. There is a family of *t* distributions, depending on the number of degrees of freedom.

IV. A proportion is a ratio, fraction, or percent that indicates the part of the sample or population that has a particular characteristic.

A. A sample proportion, *p*, is found by *x*, the number of successes, divided by *n*, the number of observations.

B. We construct a confidence interval for a sample proportion from the following formula.

$$p \pm z \sqrt{\frac{p(1-p)}{n}} \qquad \textbf{(9–4)}$$

V. We can determine an appropriate sample size for estimating both means and proportions.

A. There are three factors that determine the sample size when we wish to estimate the mean.

1. The margin of error, *E*.

2. The desired level of confidence.

3. The variation in the population.

4. The formula to determine the sample size for the mean is

$$n = \left(\frac{z\sigma}{E}\right)^2 \qquad \textbf{(9–5)}$$

B. There are three factors that determine the sample size when we wish to estimate a proportion.

1. The margin of error, *E*.

2. The desired level of confidence.

3. A value for π to calculate the variation in the population.

4. The formula to determine the sample size for a proportion is

$$n = \pi(1-\pi)\left(\frac{z}{E}\right)^2 \qquad \textbf{(9–6)}$$

VI. For a finite population, the standard error is adjusted by the factor: $\sqrt{\frac{N-n}{N-1}}$

CHAPTER EXERCISES

31. A random sample of 85 group leaders, supervisors, and similar personnel at General Motors revealed that, on average, they spent 6.5 years in a particular job before being promoted. The standard deviation of the sample was 1.7 years. Construct a 95% confidence interval.

32. A state meat inspector in Iowa has been given the assignment of estimating the mean net weight of packages of ground chuck labeled "3 pounds." Of course, he realizes that the weights cannot always be precisely 3 pounds. A sample of 36 packages reveals the mean weight to be 3.01 pounds, with a standard deviation of 0.03 pound.

a. What is the estimated population mean?
b. Determine a 95% confidence interval for the population mean.

33. As part of their business promotional package, the Milwaukee Chamber of Commerce would like an estimate of the mean cost per month to lease a one-bedroom apartment. The mean cost per month for a random sample of 40 apartments currently available for lease was \$884. The standard deviation of the sample was \$50.

a. Develop a 98% confidence interval for the population mean.
b. Would it be reasonable to conclude that the population mean is \$950 per month?

34. A recent survey of 50 executives who were laid off during a recent recession revealed it took a mean of 26 weeks for them to find another position. The standard deviation of the sample was 6.2 weeks. Construct a 95% confidence interval for the population mean. Is it reasonable that the population mean is 28 weeks? Justify your answer.

35. Marty Rowatti recently assumed the position of director of the YMCA of South Jersey. He would like some current data on how long current members of the YMCA have been members. To investigate, suppose he selects a random sample of 40 current members. The mean length of membership for the sample is 8.32 years and the standard deviation is 3.07 years.

a. What is the mean of the population?
b. Develop a 90% confidence interval for the population mean.
c. The previous director, in the summary report she prepared as she retired, indicated the mean length of membership was now "almost 10 years." Does the sample information substantiate this claim? Cite evidence.

36. The American Restaurant Association collected information on the number of meals eaten outside the home per week by young married couples. A survey of 60 couples showed the sample mean number of meals eaten outside the home was 2.76 meals per week, with a standard deviation of 0.75 meal per week. Construct a 99% confidence interval for the population mean.

37. The National Collegiate Athletic Association (NCAA) reported that college football assistant coaches spend a mean of 70 hours per week on coaching and recruiting during the season. A random sample of 50 assistant coaches showed the sample mean to be 68.6 hours, with a standard deviation of 8.2 hours.

a. Using the sample data, construct a 99% confidence interval for the population mean.
b. Does the 99% confidence interval include the value suggested by the NCAA? Interpret this result.
c. Suppose you decided to switch from a 99% to a 95% confidence interval. Without performing any calculations, will the interval increase, decrease, or stay the same? Which of the values in the formula will change?

38. The Human Relations Department of Electronics Inc. would like to include a dental plan as part of the benefits package. The question is: How much does a typical employee and his or her family spend per year on dental expenses? A sample of 45 employees reveals the mean amount spent last year was \$1,820, with a standard deviation of \$660.

a. Construct a 95% confidence interval for the population mean.
b. The information from part (a) was given to the president of Electronics Inc. He indicated he could afford \$1,700 of dental expenses per employee. Is it possible that the population mean could be \$1,700? Justify your answer.

39. A student conducted a study and reported that the 95% confidence interval for the mean ranged from 46 to 54. He was sure that the mean of the sample was 50, that the standard deviation of the sample was 16, and that the sample size was at least 30, but could not remember the exact number. Can you help him out?

40. A recent study by the American Automobile Dealers Association surveyed a random sample of 20 dealers. The data revealed a mean amount of profit per car sold was \$290, with a standard deviation of \$125. Develop a 95% confidence interval for the population mean of profit per car.

41. A study of 25 graduates of four-year public colleges revealed the mean amount owed by a student in student loans was \$55,051. The standard deviation of the sample was \$7,568. Construct a 90% confidence interval for the population mean. Is it reasonable to conclude that the mean of the population is actually \$55,000? Explain why or why not.

42. An important factor in selling a residential property is the number of times real estate agents show a home. A sample of 15 homes recently sold in the Buffalo, New York, area revealed the mean number of times a home was shown was 24 and the standard deviation of the sample was 5 people. Develop a 98% confidence interval for the population mean.

43. FILE In 2003, the Accreditation Council for Graduate Medical Education (ACGME) implemented new rules limiting work hours for all residents. A key component of these rules is that residents should work no more than 80 hours per week. The following is the number of weekly hours worked in 2017 by a sample of residents at the Tidelands Medical Center.

84	86	84	86	79	82	87	81	84	78	74	86

a. What is the point estimate of the population mean for the number of weekly hours worked at the Tidelands Medical Center?
b. Develop a 90% confidence interval for the population mean.
c. Is the Tidelands Medical Center within the ACGME guideline? Why?

44. FILE PrintTech, Inc. is introducing a new line of ink-jet printers and would like to promote the number of pages a user can expect from a print cartridge. A sample of 10 cartridges revealed the following number of pages printed.

2,698	2,028	2,474	2,395	2,372	2,475	1,927	3,006	2,334	2,379

a. What is the point estimate of the population mean?
b. Develop a 95% confidence interval for the population mean.

45. FILE Dr. Susan Benner is an industrial psychologist. She is currently studying stress among executives of Internet companies. She has developed a questionnaire that she believes measures stress. A score above 80 indicates stress at a dangerous level. A random sample of 15 executives revealed the following stress level scores.

94	78	83	90	78	99	97	90	97	90	93	94	100	75	84

a. Find the mean stress level for this sample. What is the point estimate of the population mean?
b. Construct a 95% confidence level for the population mean.
c. According to Dr. Benner's test, is it reasonable to conclude that the mean stress level of Internet executives is 80? Explain.

46. Pharmaceutical companies promote their prescription drugs using television advertising. In a survey of 80 randomly sampled television viewers, 10 indicated that they asked their physician about using a prescription drug they saw advertised on TV. Develop a 95% confidence interval for the proportion of viewers who discussed a drug seen on TV with their physician. Is it reasonable to conclude that 25% of the viewers discuss an advertised drug with their physician?

47. HighTech, Inc. randomly tests its employees about company policies. Last year in the 400 random tests conducted, 14 employees failed the test. Develop a 99% confidence interval for the proportion of applicants that fail the test. Would it be reasonable to conclude that 5% of the employees cannot pass the company policy test? Explain.

48. During a national debate on changes to health care, a cable news service performs an opinion poll of 500 small-business owners. It shows that 65% of small-business owners do not approve of the changes. Develop a 95% confidence interval for the proportion opposing health care changes. Comment on the result.

49. There are 20,000 eligible voters in York County, South Carolina. A random sample of 500 York County voters revealed 350 plan to vote to return Louella Miller to the state senate. Construct a 99% confidence interval for the proportion of voters in the county who plan to vote for Ms. Miller. From this sample information, is it reasonable to conclude that Ms. Miller will receive a majority of the votes?

50. In a poll to estimate presidential popularity, each person in a random sample of 1,000 voters was asked to agree with one of the following statements:

1. The president is doing a good job.
2. The president is doing a poor job.
3. I have no opinion.

A total of 560 respondents selected the first statement, indicating they thought the president was doing a good job.

a. Construct a 95% confidence interval for the proportion of respondents who feel the president is doing a good job.
b. Based on your interval in part (a), is it reasonable to conclude that a majority of the population believes the president is doing a good job?

51. Police Chief Edward Wilkin of River City reports 500 traffic citations were issued last month. A sample of 35 of these citations showed the mean amount of the fine was $54, with a standard deviation of $4.50. Construct a 95% confidence interval for the mean amount of a citation in River City.

52. The First National Bank of Wilson has 650 checking account customers. A recent sample of 50 of these customers showed 26 have a Visa card with the bank. Construct the 99% confidence interval for the proportion of checking account customers who have a Visa card with the bank.

53. It is estimated that 60% of U.S. households subscribe to cable TV. You would like to verify this statement for your class in mass communications. If you want your estimate to be within 5 percentage points, with a 95% level of confidence, how many households should you sample?

54. You need to estimate the mean number of travel days per year for salespeople. The mean of a small pilot study was 150 days, with a standard deviation of 14 days. If you must estimate the population mean within 2 days, how many salespeople should you sample? Use the 90% confidence level.

55. You want to estimate the mean family income in a rural area of central Indiana. The question is, how many families should be sampled? In a pilot sample of 10 families, the standard deviation of the sample was $500. The sponsor of the survey wants you to use the 95% confidence level. The estimate is to be within $100. How many families should be interviewed?

56. *Families USA,* a monthly magazine that discusses issues related to health and health costs, surveyed 20 of its subscribers. It found that the annual health insurance premiums for a family with coverage through an employer averaged $10,979. The standard deviation of the sample was $1,000.

a. Based on this sample information, develop a 90% confidence interval for the population mean yearly premium.
b. How large a sample is needed to find the population mean within $250 at 99% confidence?

57. Passenger comfort is influenced by the amount of pressurization in an airline cabin. Higher pressurization permits a closer-to-normal environment and a more relaxed flight. A study by an airline user group recorded the equivalent air pressure on 30 randomly chosen flights. The study revealed a mean equivalent air pressure of 8,000 feet with a standard deviation of 300 feet.

a. Develop a 99% confidence interval for the population mean equivalent air pressure.
b. How large a sample is needed to find the population mean within 25 feet at 95% confidence?

58. A survey of 25 randomly sampled judges employed by the state of Florida found that they earned an average wage (including benefits) of $65.00 per hour. The sample standard deviation was $6.25 per hour.
   a. What is the population mean? What is the best estimate of the population mean?
   b. Develop a 99% confidence interval for the population mean wage (including benefits) for these employees.
   c. How large a sample is needed to assess the population mean with an allowable error of $1.00 at 95% confidence?

59. Based on a sample of 50 U.S. citizens, the American Film Institute found that a typical American spent 78 hours watching movies last year. The standard deviation of this sample was 9 hours.
   a. Develop a 95% confidence interval for the population mean number of hours spent watching movies last year.
   b. How large a sample should be used to be 90% confident the sample mean is within 1.0 hour of the population mean?

60. Dylan Jones kept careful records of the fuel efficiency of his new car. After the first nine times he filled up the tank, he found the mean was 23.4 miles per gallon (mpg) with a sample standard deviation of 0.9 mpg.
   a. Compute the 95% confidence interval for his mpg.
   b. How many times should he fill his gas tank to obtain a margin of error below 0.1 mpg?

61. A survey of 36 randomly selected iPhone owners showed that the purchase price has a mean of $650 with a sample standard deviation of $24.
   a. Compute the standard error of the sample mean.
   b. Compute the 95% confidence interval for the mean.
   c. How large a sample is needed to estimate the population mean within $10?

62. You plan to conduct a survey to find what proportion of the workforce has two or more jobs. You decide on the 95% confidence level and a margin of error of 2%. A pilot survey reveals that 5 of the 50 sampled hold two or more jobs. How many in the workforce should be interviewed to meet your requirements?

63. A study conducted several years ago reported that 21 percent of public accountants changed companies within 3 years. The American Institute of CPA's would like to update the study. They would like to estimate the population proportion of public accountants who changed companies within 3 years with a margin of error of 3% and a 95% level of confidence.
   a. To update this study, the files of how many public accountants should be studied?
   b. How many public accountants should be contacted if no previous estimates of the population proportion are available?

64. As part of an annual review of its accounts, a discount brokerage selected a random sample of 36 customers and reviewed the value of their accounts. The mean was $32,000 with a sample standard deviation of $8,200. What is a 90% confidence interval for the mean account value of the population of customers?

65. The National Weight Control Registry tries to mine secrets of success from people who lost at least 30 pounds and kept it off for at least a year. It reports that out of 2,700 registrants, 459 were on a low-carbohydrate diet (less than 90 grams a day).
   a. Develop a 95% confidence interval for the proportion of people on a low-carbohydrate diet.
   b. Is it possible that the population percentage is 18%?
   c. How large a sample is needed to estimate the proportion within 0.5%?

66. Near the time of an election, a cable news service performs an opinion poll of 1,000 probable voters. It shows that the Republican contender has an advantage of 52% to 48%.
   a. Develop a 95% confidence interval for the proportion favoring the Republican candidate.
   b. Estimate the probability that the Democratic candidate is actually leading.
   c. Repeat the above analysis based on a sample of 3,000 probable voters.

67. A sample of 352 subscribers to *Wired* magazine shows the mean time spent using the Internet is 13.4 hours per week, with a sample standard deviation of 6.8 hours. Find the 95% confidence interval for the mean time *Wired* subscribers spend on the Internet.

68. The Tennessee Tourism Institute (TTI) plans to sample information center visitors entering the state to learn the fraction of visitors who plan to camp in the state. Current estimates are that 35% of visitors are campers. How many visitors would you sample to estimate the population proportion of campers with a 95% confidence level and an allowable error of 2%?

DATA ANALYTICS

69. **FILE** Refer to the North Valley Real Estate data, which reports information on homes sold in the area during the last year. Select a random sample of twenty homes.
- **a.** Based on your random sample of twenty homes, develop a 95% confidence interval for the mean selling price of the homes.
- **b.** Based on your random sample of twenty homes, develop a 95% confidence interval for the mean days on the market.
- **c.** Based on your random sample of twenty homes, develop a 95% confidence interval for the proportion of homes with a pool.
- **d.** Suppose that North Valley Real Estate employs several agents. Each agent will be randomly assigned twenty homes to sell. The agents are highly motivated to sell homes based on the commissions they earn. They are also concerned about the twenty homes they are assigned to sell. Using the confidence intervals you created, write a general memo informing the agents about the characteristics of the homes they may be assigned to sell.
- **e.** What would you do if your confidence intervals did not include the mean of all 105 homes? How could this happen?

70. **FILE** Refer to the Baseball 2016 data, which report information on the 30 Major League Baseball teams for the 2016 season. Assume the 2016 data represents a sample.
- **a.** Develop a 95% confidence interval for the mean number of home runs per team.
- **b.** Develop a 95% confidence interval for the mean batting average by each team.
- **c.** Develop a 95% confidence interval for the mean earned run average (ERA) for each team.

71. **FILE** Refer to the Lincolnville School District bus data.
- **a.** Develop a 95% confidence interval for the mean bus maintenance cost.
- **b.** Develop a 95% confidence interval for the mean bus odometer miles.
- **c.** Write a business memo to the state transportation official to report your results.

A REVIEW OF CHAPTERS 8–9

We began Chapter 8 by describing the reasons sampling is necessary. We sample because it is often impossible to study every item, or individual, in some populations. For example, to contact all U.S. bank officers and record their annual incomes would be too expensive and time-consuming, . Also, sampling often destroys the product. A drug manufacturer cannot test the properties of each vitamin tablet manufactured because there would be none left to sell. Therefore, to estimate a population parameter, we select a sample from the population. A sample is a part of the population. Care must be taken to ensure that every member of our population has a chance of being selected; otherwise, the conclusions might be biased. A number of probability-type sampling methods can be used, including *simple random, systematic, stratified,* and *cluster sampling.*

Regardless of the sampling method selected, a sample statistic is seldom equal to the corresponding population parameter. For example, the mean of a sample is seldom exactly the same as the mean of the population. The difference between this sample statistic and the population parameter is the *sampling error.*

In Chapter 8, we demonstrated that, if we select all possible samples of a specified size from a population and calculate the mean of these samples, the result will be exactly equal to the population mean. We also showed that the dispersion in the distribution of the sample means is equal to the population standard deviation divided by the square root of the sample size. This result is called the standard error of the mean. There is less dispersion in the distribution of the sample means than in the population. In addition, as we increase the number of observations in each sample, we decrease the variation in the sampling distribution.

The central limit theorem is the foundation of statistical inference. It states that, if the population from which we select the samples follows the normal probability distribution, the distribution of the sample means will also follow the normal distribution. If the population is not normal, it will approach the normal probability distribution as we increase the size of the sample.

Our focus in Chapter 9 was point estimates and interval estimates. A point estimate is a single value used to estimate a population parameter. An interval estimate is a range of values within which we expect the population parameter to occur. For example, based on a sample, we estimate that the mean annual income of all professional house painters in Atlanta, Georgia (the population), is $45,300. That estimate is called a *point estimate.* If we state that the population mean is probably in the interval between $45,200 and $45,400, that estimate is called an *interval estimate.* The two endpoints ($45,200 and $45,400) are the *confidence limits* for the population mean. We also described procedures for establishing a confidence interval for a population mean when the population standard deviation is not known and for a population proportion. In this chapter, we also provided a method to determine the necessary sample size based on the dispersion in the population, the level of confidence desired, and the desired precision of the estimate or margin of error.

PROBLEMS

1. A recent study indicated that women took an average of 8.6 weeks of unpaid leave from their jobs after the birth of a child. Assume that this distribution follows the normal probability distribution with a standard deviation of 2.0 weeks. We select a sample of 35 women who recently returned to work after the birth of a child. What is the likelihood that the mean of this sample is at least 8.8 weeks?
2. The manager of Tee Shirt Emporium reports that the mean number of shirts sold per week is 1,210, with a standard deviation of 325. The distribution of sales follows the normal distribution. What is the likelihood of selecting a sample of 25 weeks and finding the sample mean to be 1,100 or less?
3. The owner of the Gulf Stream Café wished to estimate the mean number of lunch customers per day. A sample of 40 days revealed a mean of 160 per day, with a standard deviation of 20 per day. Develop a 98% confidence interval for the mean number of customers per day.
4. The manager of the local Hamburger Express wishes to estimate the mean time customers spend at the drive-through window. A sample of 20 customers experienced a mean waiting time of 2.65 minutes, with a standard deviation of 0.45 minute. Develop a 90% confidence interval for the mean waiting time.
5. Defiance Tool and Die has 293 sales offices throughout the world. The VP of Sales is studying the usage of its copy machines. A random sample of six of the sales offices revealed the following number of copies made in each selected office last week.

826	931	1,126	918	1,011	1,101

   Develop a 95% confidence interval for the mean number of copies per week.
6. John Kleman is the host of KXYZ Radio 55 AM drive-time news in Denver. During his morning program, John asks listeners to call in and discuss current local and national news. This morning, John was concerned with the number of hours children under 12 years of age watch TV per day. The last five callers reported that their children watched the following number of hours of TV last night.

3.0	3.5	4.0	4.5	3.0

   Would it be reasonable to develop a confidence interval from these data to show the mean number of hours of TV watched? If yes, construct an appropriate confidence interval and interpret the result. If no, why would a confidence interval not be appropriate?
7. Historically, Widgets Manufacturing Inc. produces 250 widgets per day. Recently the new owner bought a new machine to produce more widgets per day. A sample of 16 days' production revealed a mean of 240 units with a standard deviation of 35. Construct a confidence interval for the mean number of widgets produced per day. Does it seem reasonable to conclude that the mean daily widget production has changed? Justify your conclusion.
8. A manufacturer of cell phone batteries wants to estimate the useful life of its battery (in thousands of hours). The estimate is to be within 0.10 (100 hours). Assume a 95% level of confidence and that the standard deviation of the useful life of the battery is 0.90 (900 hours). Determine the required sample size.
9. The manager of a home improvement store wishes to estimate the mean amount of money spent in the store. The estimate is to be within $4.00 with a 95% level of confidence. The manager does not know the standard deviation of the amounts spent. However, he does estimate that the range is from $5.00 up to $155.00. How large of a sample is needed?
10. In a sample of 200 residents of Georgetown County, 120 reported they believed the county real estate taxes were too high. Develop a 95% confidence interval for the proportion of residents who believe the tax rate is too high. Does it seem reasonable to conclude that 50% of the voters believe that taxes are too high?
11. In recent times, the percent of buyers purchasing a new vehicle via the Internet has been large enough that local automobile dealers are concerned about its impact on their business. The information needed is an estimate of the proportion of purchases via the Internet. How large of a sample of purchasers is necessary for the estimate to be

within 2 percentage points with a 98% level of confidence? Current thinking is that about 8% of the vehicles are purchased via the Internet.

12. Historically, the proportion of adults over the age of 24 who smoke has been .30. In recent years, much information has been published and aired on radio and TV that smoking is not good for one's health. A sample of 500 adults revealed only 25% of those sampled smoked. Develop a 98% confidence interval for the proportion of adults who currently smoke. Does it seem reasonable to conclude that the proportion of adults who smoke has changed?

13. The auditor of the state of Ohio needs an estimate of the proportion of residents who regularly play the state lottery. Historically, about 40% regularly play, but the auditor would like some current information. How large a sample is necessary for the estimate to be within 3 percentage points, with a 98% level of confidence?

CASES

Century National Bank

Refer to the description of Century National Bank at the end of the Review of Chapters 1–4 on page 129. When Mr. Selig took over as president of Century several years ago, the use of debit cards was just beginning. He would like an update on the use of these cards. Develop a 95% confidence interval for the proportion of customers using these cards. On the basis of the confidence interval, is it reasonable to conclude that more than half of the customers use a debit card? Write a brief report interpreting the results.

PRACTICE TEST

Part 1—Objective

1. If each item in the population has the same chance of being selected, this is called a ________. **1.** ____________
2. The difference between the population mean and the sample mean is called the ________. **2.** ____________
3. The ________ is the standard deviation of the distribution of sample means. **3.** ____________
4. If the sample size is increased, the variance of the sample means will ________. (become smaller, become larger, not change) **4.** ____________
5. A single value used to estimate a population parameter is called a ________. **5.** ____________
6. A range of values within which the population parameter is expected to occur is called a ________. **6.** ____________
7. Which of the following does *not* affect the width of a confidence interval? (sample size, variation in the population, level of confidence, size of population) **7.** ____________
8. The fraction of a population that has a particular characteristic is called a ________. **8.** ____________
9. Which of the following is not a characteristic of the *t* distribution? (positively skewed, continuous, mean of zero, based on degrees of freedom) **9.** ____________
10. To determine the required sample size of a proportion when no estimate of the population proportion is available, what value is used? **10.** ____________

Part 2—Problems

1. Americans spend an average (mean) of 12.2 minutes (per day) in the shower. The distribution of times follows the normal distribution with a population standard deviation of 2.3 minutes. What is the likelihood that the mean time per day for a sample of 12 Americans was 11 minutes or less?
2. A recent study of 26 Conway, South Carolina, residents revealed they had lived at their current address an average of 9.3 years. The standard deviation of the sample was 2 years.
   a. What is the population mean?
   b. What is the best estimate of the population mean?
   c. What is the standard error of estimate?
   d. Develop a 90% confidence interval for the population mean.
3. A recent federal report indicated that 27% of children ages 2 to 5 ate a vegetable at least five times a week. How large a sample is needed to estimate the true population proportion within 2% with a 98% level of confidence? Be sure to use the information contained in the federal report.
4. The Philadelphia Area Transit Authority wishes to estimate the proportion of central city workers that use public transportation to get to work. A sample of 100 workers revealed that 64 used public transportation. Develop a 95% confidence interval for the population proportion.